# SRS/SDD Self-Review Report – Review 2

## 1. Run Information

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|---|---|
| **Project code** | FA26SE036 |
| **Project name** | Develop an AI-Assisted Omnichannel Order Fulfillment and Packaging Optimization System (OptiPack AI) |
| **Team** | OptiPack AI – Nguyễn Phương Mỹ Thuận (leader), Lê Đức Trung Thi, Huỳnh Quốc Việt, Phan Huỳnh Hải Phượng |
| **Supervisor** | Thân Thị Ngọc Vân |
| **AI tool and model** | Claude (Cowork mode, Claude desktop app) with read access to the whole repository; model claude-opus-5-5 |
| **Run date** | 25/10/2026 |
| **Repository tag / commit** | https://github.com/ThuaanNguyeen111/OptiPackAI – tag review-2, commit 1a3101cb930983c2a643605740e8a9752042c00b |
| **Registration form version used** | Version 1.1 (proposed final, 25/10/2026) with Form BM03 |
| **Version** | 1.0 |
| **Date** | 25/10/2026 |

## 2. Standard Prompt

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You are reviewing the SRS and SDD of a software capstone project against its
registration form and source code. Report only; do not change any file.

Inputs: the registration form, SRS, SDD (with diagrams), source code and
database script in this repository.

Check:
1. Scope: FRs in the SRS not covered by the registration form, and
   registration items with no FR.
2. Requirements: FRs without an actor or testable acceptance criteria;
   NFRs without metric, threshold and measurement method.
3. Data: entities missing from the Logical ERD; ERD entities or attributes
   missing from the database script, and tables not in the ERD.
4. States: for each main entity, states used in code but missing from its
   state machine; transitions with no triggering FR; states with no exit.
5. Flows: main business flows in activity diagrams with no implementation
   or no test.
6. Architecture: components or third-party services in code but not in the
   architecture diagram, and the reverse.
7. Design vs code: modules, APIs or business rules implemented differently
   from the SDD.

Output one table: ID | Category | Location | Finding | Severity
(High/Medium/Low) | Evidence. Then list the 5 most important findings.
If something cannot be verified from the repository, say so; do not guess.
```

## 3. Findings and Team Response

| ID | Category | Location | Finding | Severity | Evidence | Team response |
|---|---|---|---|---|---|---|
| F01 | Scope | Registration v1.1 §3.2 "Products – Mobile Application"; SRS §4.1; SDD | The registration still lists a Mobile Application (warehouse picking, packing confirmation, barcode/QR scanning, shipment tracking, push | High | Registration v1.1 "Mobile Application" bullets; SRS 4.1 "a mobile scanning app | Planned – Việt, 20/11/2026: build the Flutter mobile app in Iteration 3 (Design Brief G1). |

| ID | Category | Location | Finding | Severity | Evidence | Team response |
|---|---|---|---|---|---|---|
| | | §1.1 | notifications) but the SRS has no FR for it; SRS and SDD only state "planned for Iteration 3". No mobile source code exists in the repository. | | is planned for Iteration 3 and is not part of this release"; SDD 1.1 component row "Mobile app – Flutter (planned)"; repository tree has only be/, fe/, storefront/, packer/. | |
| F02 | Scope | Registration v1.1 §3.1 "Order Consolidation – Merge orders from different platforms"; SRS BR-01, BR-03, §3.3.2; code | The registration requires merging orders from different platforms, but the SRS limits grouping to the same shop and the code deliberately puts platform into the consolidation key, so orders of different platforms are never merged. No FR covers cross-platform merging. | High | SRS BR-01 "Orders of the same shop with the same consolidation key…", BR-03 "merge requires same shop"; be/src/modules/orders/utils/consolidation-key.util.ts lines 70–80 ("platform BẮT BUỘC có mặt trong key"). | Kept – each platform has its own carrier order, so orders of different platforms stay in separate groups; same-recipient groups can be linked (UC-16). Wording to be confirmed with the reviewers at Review 2 (Design Brief G6). |
| F03 | States | SDD §3.8; SDD §3 workflow table | Only the Order Group has a state machine diagram. Packing Plan, Shipment, Return Request and Order have lifecycles in code and in the SDD workflow table but no state machine diagram. | High | SDD 3.8 contains only "Figure 11. State machine diagram – Order Group"; be/src/modules/packing/schemas/packing-plan.schema.ts lines 21–31 (PACKING_PLAN_STATUSES); be/src/modules/shipments/enums/shipment-status.enum.ts; be/src/modules/shipments/enums/return.enums.ts; be/src/modules/orders/enums/order-status.enum.ts. | Planned – Thuận: Order state machine added 08/11/2026 (SDD Figure 13, draw.io); Packing Plan, Shipment and Return Request still to draw (Design Brief G7). |

| ID | Category | Location | Finding | Severity | Evidence | Team response |
|---|---|---|---|---|---|---|
| F04 | Architecture | SDD Figure 1; docs/diagrams/architecture/general_architecture_diagram.puml | The architecture diagram shows a "Mobile App (Flutter)" container that does not exist in code, and omits components and services that do exist: the AURELLE website (Next.js, storefront/), the AURELLE Open API simulator (be/scripts/aurelle-mock-server.ts), Google OAuth and Gmail SMTP. The SDD text (§1.1) describes two web front ends and the simulator, so diagram and text disagree. | High | general_architecture_diagram.puml line 14 Container(FE_APP, "Mobile App", "Flutter"); no Next.js / storefront container; storefront/app/; be/scripts/aurelle-mock-server.ts; be/src/modules/auth/auth.service.ts (Google userinfo); be/src/modules/mail/mail.service.ts. | Planned – Thuận: architecture redrawn 08/11/2026 in draw.io (SDD Figure 1) with HTTPS links and the mobile app marked as planned in SDD 1.1; Google OAuth and SMTP still to add to the figure (Design Brief G8). |
| F05 | Data | SRS Figure 8 / SDD Figure 3 (Logical ERD); SRS §3.1.5 entity table; be/src//schemas | The Logical ERD (22 entities) is missing many main entities that are listed in the SRS entity table and exist as collections: Stock Reservation, Pick Event, Warehouse / Zone / Bin Location, SKU Bin Assignment, Inventory Movement, Category / Colour / Master SKU, Marketplace SKU Mapping, Product Master, Shipment Event, Notification, Activity Log, Status Outbox, Packing Settings. Staff roles (Store Owner, Warehouse Staff, Packaging Staff, Shipping Coordinator) are drawn as separate entities while the code has a single users collection with a role; "Website" has no collection. | High | ERD image docs/diagrams/erd/export/erd.png and erd.puml; SRS 3.1.5 rows 1–20; collections users, stock_reservations, pick_events, bin_locations, sku_bin_assignments, inventory_movements, master_skus, marketplace_sku_mappings, product_master, shipment_events, notifications, activity_logs, marketplace_status_outbox. | Planned – Thi, 15/11/2026: extend the Logical ERD to all main entities (Design Brief G9). |
| F06 | Data | Repository (database script) | No database script exists in the repository (collections are defined only by Mongoose schemas; be/scripts holds seed and migration scripts). Therefore "ERD entities or attributes missing from the database script" cannot be | Medium | git grep "collection: '" under be/src → 57 distinct collections; be/scripts/ contains seed-*.ts, migrate-*.ts, backfill- | Planned – Thi, 15/11/2026: add a database script (collections and indexes) matching the ERD (Design Brief G9). |

| ID | Category | Location | Finding | Severity | Evidence | Team response |
|---|---|---|---|---|---|---|
| | | | verified. The code defines 57 collections versus 22 entities in the ERD. | | *.ts only. | |
| F07 | Scope | Registration v1.1 §3.1 "Order Consolidation – Group orders by warehouse"; "Warehouse Fulfillment – Batch Picking" | No FR in the SRS for grouping orders by warehouse or for batch picking across several order groups; picking is defined per order group (SRS 3.5.1). The code has a picking-route endpoint described as "wave picking … (Mobile App)" that the SRS does not mention. | Medium | SRS 3.3.2 (key = platform + phone + address + city), 3.5.1; be/src/modules/warehouse/warehouse.controller.ts line 471. | Kept – picking is done per order group by design (stock is reserved per group; orders of the same recipient are already grouped and a group belongs to one warehouse). The picking-route endpoint is kept for the mobile app planned in Iteration 3. |
| F08 | Scope | Registration v1.1 §3.1 "Shipping Cost Estimation" (volumetric weight, packaging cost, total fulfillment cost; recommend courier, delivery method, expected delivery time) | The SRS has no explicit FR or acceptance criteria for volumetric weight, packaging cost, total fulfillment cost or expected delivery time; it only states "compare quotes, choose carrier and service" (UC-37). The code contains GHN lead time, converted weight and packing cost objective, which are not specified. | Medium | SRS UC-37, 3.7.1; be/src/modules/carriers/adapters/ghn.adapter.ts (converted_weight, lead time), be/src/modules/packing/solver/objective.ts. | Fixed 08/11/2026 – SRS 3.7.1 now specifies chargeable / volumetric weight, total fulfilment cost (shipping + box + fillers), carrier strategy, expected delivery time and an acceptance criterion. |
| F09 | Scope | SRS UC-45–UC-51, UC-42–UC-44, UC-24 | SRS FRs that are not itemised in the registration form: AURELLE customer features (register, cart, checkout, cancel, return request), AURELLE web admin and developer apps, return / exchange processing with inspection and replacement orders, stock receipts by lot (FIFO). The registration mentions only AURELLE as a sync channel and "Track Returns". | Medium | Registration v1.1 §3.1 lists AURELLE only under synchronization; SRS 2.2.2 rows UC-24, UC-42–UC-51. | Kept – these FRs implement the AURELLE channel and return handling accepted in registration v1.1 (Form BM03 #1); no scope change. |
| F10 | Scope | Registration v1.1 actors (Store Owner "View business dashboards", "Analyze operational reports"); SRS §3.1.2 row 6 | The Dashboard Overview required for the Store Owner is only partly implemented: revenue-by-channel, finance and staff widgets show sample data (stated in the SRS). | Medium | SRS 3.1.2 row 6; fe/src/pages/DashboardPage.tsx, fe/src/data/dashboard-mock.ts. | Fixed 08/11/2026 – the home dashboard reads GET /dashboard/summary; sample data removed (SRS 3.1.2 row 6, Design Brief G2). |
| F11 | Requireme | SRS §4.2.1 | Usability NFRs have no | Medium | SRS 4.2.1: | Fixed 08/11/2026 – SRS |

| ID | Category | Location | Finding | Severity | Evidence | Team response |
|---|---|---|---|---|---|---|
| | nts | Usability | metric, threshold or measurement method. | | three statements without metric / threshold / method. | 4.2.1 now has a task-time metric (≤ 5 minutes, 4 of 5 users) and a measurement method. |
| F12 | States | SDD §3.8 Figure 11; docs/diagrams/state/state_diagram.puml; allowed-status-transitions.ts | The Order Group state machine does not show several transitions allowed in code: awaiting_packaging → partial_needs_review, picked → picking, partial_needs_review → picked, approved_for_packing → picked / picking, shipped → packed, delivered → returned. Composite-state arrows ("Dispatch / Void", "Restocked / Closed" to CANCELED) imply shipped / delivered → canceled, which the code does not allow. | Medium | be/src/modules/order-groups/enums/allowed-status-transitions.ts (33 transitions); docs/diagrams/state/state_diagram.puml (16 state-level transitions + composite arrows). | Planned – Thuận, 08/11/2026: redraw the Order Group state machine with all transitions of allowed-status-transitions.ts (with Design Brief G7). |
| F13 | States | SDD §3 workflow table WF-03 | Packing Plan states in code include rejected, failed and superseded; the SDD lists only computing → ready → approved → packing → packed. | Medium | be/src/modules/packing/schemas/packing-plan.schema.ts lines 21–31; SDD WF-03 row. | Fixed 08/11/2026 – SDD workflow table WF-03 lists rejected, failed and superseded; the Packing Plan state machine is still planned with G7. |
| F14 | Flows | WF-01 … WF-06; be/test | Each workflow has unit tests on its services, but there is no end-to-end test of any main workflow; be/test contains only app.e2e-spec.ts and auth.e2e-spec.ts. Unit tests use mocked models. | Medium | be/test/ listing; e.g. orders.service.tryConsolidate.spec.ts, order-groups.service.pickItem.spec.ts, packing-plan.service.spec.ts, shipments.service.spec.ts, returns.service.spec.ts, warehouse.service.receipt.spec.ts. | Planned – Thi, 04/12/2026: end-to-end tests for pick → pack → ship → return (Design Brief G4). |
| F15 | Architecture | general_architecture_diagram.puml line 40 | The API ↔ Optimization Worker link is labelled "gRPC / HTTP2", but the back end calls the packer with HTTP POST (axios, JSON). | Medium | be/src/modules/packing/solver/cp-sat.ts lines 1, 94 (axios.post); packer/packer | Fixed 08/11/2026 – the redrawn architecture diagram (SDD Figure 1) labels the API → packing optimizer link as HTTP. |

| ID | Category | Location | Finding | Severity | Evidence | Team response |
|---|---|---|---|---|---|---|
| | | | | | /main.py (FastAPI). | |
| F16 | Design vs code | SDD §1.2, §4.1–4.10 | The package diagram (Figure 2) and all class and sequence diagrams of the detailed design are placeholders, so the module design cannot be compared with the code beyond the package table and class name lists. | Medium | SDD 1.2 "[Insert Figure 2 – Package Diagram]"; SDD 4.x "[Insert figure – …]" (33 placeholders). | Planned – Thuận, 15/11/2026: package diagram and class / sequence diagrams of SDD section 4, each section drawn by its workflow owner. |
| F17 | Design vs code | SRS UC-31, §3.6.3, §3.8.2, §3.1.3; fe/src/lib/rbac.ts | Screen authorization is inconsistent: UC-31 / 3.6.3 give Warehouse Staff packing work and 3.8.2 says Warehouse Staff inspects on the "Returns screen", but 3.1.3 and rbac.ts give Warehouse Staff neither /app/packing nor /app/returns (inspection is done in the Returns tab of Warehouse Picking). The back end allows Warehouse Staff on some packing-session and return endpoints. | Medium | fe/src/lib/rbac.ts lines 9–19; fe/src/pages/WarehousePage.tsx line 151; be/src/modules/packing/packing-plan.controller.ts lines 539, 657; be/src/modules/shipments/returns.controller.ts lines 98–105, 171–178. | Fixed 08/11/2026 – SRS 3.6.3 and 3.8.2 now follow the screen authorization in 3.1.3 / rbac.ts (Warehouse Staff inspects in the Returns tab of Warehouse Picking; packing-session APIs allowed by the back end only). Use cases were renumbered (UC-01 to UC-108) to match the redrawn use case diagrams. |
| F18 | Requirements | SRS §4.2.4 Security | Security requirements are listed as mechanisms without a measurement method (except lockout and token lifetimes). | Low | SRS 4.2.4. | Fixed 08/11/2026 – SRS 4.2.4 now has a measurement method (auth unit tests in CI, 403 check per role, lockout check). |
| F19 | Requirements | SRS UC-44 | Actor "System" is not one of the defined actors (§2.1 defines "System Scheduler"). | Low | SRS 2.1 actor list; UC-44 row. | Fixed 08/11/2026 – the replacement order use case (now UC-89 "Create replacement order") has actor Store Owner, as in the use case diagram. |
| F20 | Requirements | SRS §3.4.3 | Screen layout of the "Packaging profile" function is still a placeholder. | Low | SRS 3.4.3 "[Insert screenshot of the "Packaging profile" screen]". | Planned – Phượng, 08/11/2026: add the Packaging profile screenshot to SRS 3.4.3. |
| F21 | States | SDD WF-04 row | Shipment state cancelled (carrier order cancelled before pickup, SRS UC-38) is not listed in the shipment state path. | Low | be/src/modules/shipments/enums/shipment-status.enum.ts (CANCELLED = 'cancelled'); SDD WF-04 row. | Fixed 08/11/2026 – SDD workflow table WF-04 includes cancelled (carrier order cancelled before pickup); the Shipment state machine is still planned with G7. |

| ID | Category | Location | Finding | Severity | Evidence | Team response |
|---|---|---|---|---|---|---|
| F22 | Architecture | general_architecture_diagram.puml line 47 | The API ↔ AI Services link is labelled "HTTPS / SSE", but the AI call is a single non-streaming chat-completions POST. | Low | be/src/modules/packaging/packing-guide-ai.service.ts (axios.post(${baseUrl}/chat/completions), no stream). | Fixed 08/11/2026 – the redrawn architecture diagram labels the AI link as HTTPS. |
| F23 | Data | SDD §2 collection table | Collections in code that are neither in the ERD nor named in the SDD table: stock_reservation_totals, packaging_recommendations (legacy), order_sync_settings, packing_settings, shipping_carriers, shipping_settings, carrier_webhook_events. | Low | Collection list from be/src//schemas; SDD 2 rows 01–21. | Planned – Thi, 15/11/2026: list these collections in the ERD and SDD section 2 (with G9). |
| F24 | Design vs code | SDD §1.3 b "Diagrams" row; docs/ERD.md | The SDD says the ERD is Mermaid in docs/ERD.md, but the ERD used in the reports is PlantUML docs/diagrams/erd/erd.puml; docs/ERD.md states 50 collections while the code has 57. | Low | SDD 1.3 b; docs/ERD.md line 3 ("50 collection"); docs/diagrams/erd/erd.puml. | Fixed 08/11/2026 – SDD 1.3 b Diagrams row now names draw.io; docs/ERD.md (developer reference) is refreshed with G9. |
| F25 | Design vs code | SRS §4.2.3 | The measurement method cites scripts/solver-benchmark.ts; the file is be/scripts/solver-benchmark.ts. | Low | be/scripts/solver-benchmark.ts. | Fixed 08/11/2026 – SRS 4.2.3 cites be/scripts/solver-benchmark.ts. |
| F26 | Flows | SDD §3 workflow table | The "Owner" column of WF-01 … WF-06 is empty. | Low | SDD section 3 table. | Fixed 08/11/2026 – Owner column of the SDD workflow table filled (WF-01 Thuận, WF-02 Phượng, WF-03 Việt, WF-04 Thuận, WF-05 Thi, WF-06 Thi). |

## 4. Top 5 Findings (as reported by the AI)

| # | Finding ID | Summary |
|---|---|---|
| 1 | F01 | High – Mobile Application is in the registration but has no FR and no code. |
| 2 | F02 | High – Registration requires merging orders across platforms; SRS and code merge only within one shop / platform. |
| 3 | F04 | High – Architecture diagram shows a non-existent mobile app and omits the AURELLE website, Open API simulator, Google OAuth and SMTP. |
| 4 | F03 | High – State machines are missing for Packing Plan, Shipment, Return Request and Order. |
| 5 | F05 | High – Logical ERD misses most main entities of the SRS entity table and models staff roles as separate entities. |

**Checked and consistent (no finding):** scheduler intervals in SRS 3.1.4 (order sync every minute, backfill 5 min, express SLA 10 min, product master hourly + 03:00, packing auto-compute 10 s, blocked recheck 1 min, rejection reminder 15 min, shipment SLA 15 min, write-back 15 s, stock push 1 min, returns sync 2 min); write-back retry 8 attempts, back-off 30 s → 30 min; rate limit 600 requests / minute; JWT 24 h / refresh 30 days; lockout 5 attempts / 15 min; weight tolerance 20 %; AURELLE order page refresh 30 s; 57 collections; technology versions in SDD 1.3 b; Lazada adapter is read-only (packOrders exists but is not called); state names of the Order Group diagram match GroupFulfillmentStatus.

**Could not be verified from the repository:** run-time thresholds (new Lazada order visible within sync interval + 1 minute, BRKGA p95 ≤ 1 s — only the stored result be/scripts/benchmark-results/solver-v2.json was read, the benchmark was not re-run); behaviour of live external APIs (Lazada, GHN, Google, Groq); whether the automated test suite passes at this tag (not run in this review; CI run #258 for this commit is reported green on GitHub); deployment and rollback on a real cluster (no staging environment); whether UC-16 "Link order groups" works across platforms (not stated in the SRS).

## 5. Summary

| Severity | Total | Fixed | Kept | Planned |
|---|---|---|---|---|
| *High* | *5* | *0* | *1* | *4* |
| *Medium* | *12* | *6* | *2* | *4* |
| *Low* | *9* | *7* | *0* | *2* |